

Dichiarazioni

Dec : $D \rightarrow \text{nil} \mid \text{const } x: \underline{e} = e \mid D; D$
 $D \text{ in } D \mid p \in \text{Env}$
 Sintassi ↗

Semantica Statica

$\Delta \in \text{TEnv}$

$\Delta \vdash d_0 : \Delta_0$

Associa Δ_0 a $d_0 \in \text{Dec}$
 $\cap$
 TEnv

$\vdash \text{nil} : \emptyset$

$\vdash p : V$ (V è l'insieme degli id a cui dà significato p)

Associa un tipo ad ogni espressione che non contiene errori di tipo nell'uso degli operatori

$$\frac{\Delta \vdash e : \tau}{\Delta \vdash \text{const } x: \tau = e : [x \mapsto \tau]}$$

$$\frac{\Delta \vdash d_1 : \Delta_1 \quad \Delta[\Delta_1] \vdash d_2 : \Delta_2}{\Delta \vdash d_1; d_2 : \Delta_1[\Delta_2]}$$

$$\frac{\Delta \vdash d_1 : \Delta_1 \quad \Delta[\Delta_1] \vdash d_2 : \Delta_2}{\Delta \vdash d_1 \text{ in } d_2 : \Delta_2}$$

Esempio

$\Delta = \emptyset$

(non abbiamo un ambiente contestuale)

$\left\{ \begin{array}{l} \text{const } x: \text{bool} = \text{true}; d_1 \\ \text{const } x: \text{int} = 5; d_2 \\ \text{const } y: \text{int} = 6 * x; d_3 \end{array} \right.$

$d_1; \underbrace{d_2; d_3}_d$
 $d_1; d$

• $d_1; d$

per costruire l'ambiente statico di $d_1; d$
 dobbiamo costruire l'ambiente di d_1

→ $\frac{\vdash d_1 : ? \rightarrow \dots}{\vdash d_1; d : \dots}$

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① Elaboriamo d_1 staticamente

$$\frac{\vdash \text{true} : \text{bool} \quad \text{per gli assiomi delle exp}}{\vdash \text{const } x : \text{bool} = \underline{\text{true}} : [x \mapsto \text{bool}]}$$

$$\Rightarrow \frac{\emptyset \vdash d_1 : [x \mapsto \text{bool}] \quad \boxed{\emptyset [x \mapsto \text{bool}] \vdash d : ? \text{ ②}}}{\vdash d_1 ; d : ?}$$

② Elaboriamo d_2 staticamente

$$\frac{[x \mapsto \text{bool}] \vdash d_2 : ? \text{ 2a}}{\vdash d_2 ; d_3 : ?}$$

2a Elaboriamo d_2 staticamente

$$\frac{[x \mapsto \text{bool}] \vdash 5 : \text{int} \quad \text{per gli assiomi delle exp}}{[x \mapsto \text{bool}] \vdash \text{const } x : \text{int} = 5 : [x \mapsto \text{int}]}$$

$$\frac{[x \mapsto \text{bool}] \vdash d_2 : [x \mapsto \text{int}] \quad \frac{[x \mapsto \text{int}]}{[x \mapsto \text{bool}] [x \mapsto \text{int}] \vdash d_3 : ? \text{ 2b}}}{[x \mapsto \text{bool}] \vdash d_2 ; d_3 : ?}$$

2b Elaboriamo d_3 staticamente

$$\frac{[x \mapsto \text{int}] \vdash 6 * x : ? \text{ 2c}}{[x \mapsto \text{int}] \vdash \text{const } y : \text{int} = 6 * x :}$$

②c) Valutiamo $G * x$ storicamente

$$\frac{\begin{array}{c} \text{per gli assiomi} \\ \text{delle exp} \end{array} [x \mapsto \text{int}] \vdash G : \text{int} \quad \begin{array}{c} \Delta \\ [x \mapsto \text{int}] \vdash x : \text{int} \end{array} \quad \begin{array}{c} \text{per gli assiomi} \\ \text{delle exp} \end{array} \text{perché } \Delta(x) = \text{int}}{[x \mapsto \text{int}] \vdash G * x : \text{int}}$$

$$\begin{array}{c} \rightarrow \frac{[x \mapsto \text{int}] \vdash G * x : \text{int}}{[x \mapsto \text{int}] \vdash \text{const } y \text{ (if } \text{int} = G * x : [y \mapsto \text{int}])} \\ \text{chiude il passo 2b} \end{array}$$

$$\begin{array}{c} \rightarrow \frac{[x \mapsto \text{bool}] \vdash d_2 : [x \mapsto \text{int}] \quad [x \mapsto \text{int}] \vdash d_3 : [y \mapsto \text{int}]}{[x \mapsto \text{bool}] \vdash \underbrace{d_2 ; d_3}_d : \underbrace{[x \mapsto \text{int}] [y \mapsto \text{int}]}_{= [x \mapsto \text{int}, y \mapsto \text{int}]}} \end{array}$$

chiude il passo 2a
e quindi il passo 2

Di conseguenza il passo ①

$$\begin{array}{c} (*) \frac{\vdash d_1 : [x \mapsto \text{bool}] \quad [x \mapsto \text{bool}] \vdash d : [x \mapsto \text{int}, y \mapsto \text{int}]}{\vdash d_1 ; d : \underbrace{[x \mapsto \text{bool}] [x \mapsto \text{int}, y \mapsto \text{int}]}_{[x \mapsto \text{int}, y \mapsto \text{int}]}} \end{array}$$

Semantica dinamica

$$\rho \in Env \quad d, d' \in Dec \quad \rho \vdash_{\Delta} d \rightarrow_{\Delta} d'$$

$$\vdash nil \rightarrow \emptyset$$

$$\rho \in Conf. \text{terminanti}$$

$$\frac{\rho \vdash e \rightarrow^* k}{\rho \vdash \text{const } x: e = e \rightarrow [x \mapsto k]}$$

$$\frac{\rho \vdash d_1 \rightarrow d'_1}{\rho \vdash d_1; d_2 \rightarrow d'_1; d_2}$$

$$\frac{\rho[p_1] \vdash d_2 \rightarrow d'_2}{\rho \vdash p_1; d_2 \rightarrow p_1; d'_2}$$

$$\rho \vdash p_1; p_2 \rightarrow p_1[p_2]$$

$$\frac{\rho \vdash d_1 \rightarrow d'_1}{\rho \vdash d_1 \text{ in } d_2 \rightarrow d'_1 \text{ in } d_2}$$

$$\frac{\rho[p_1] \vdash d_2 \rightarrow d'_2}{\rho \vdash p_1 \text{ in } d_2 \rightarrow p_1 \text{ in } d'_2}$$

$$\rho \vdash p_1 \text{ in } p_2 \rightarrow p_2$$

Elaborare (dinamicamente) una dichiarazione

$$Elab: Dec \rightarrow Env$$

$$\forall d \in Dec. \quad Elab(d) = \rho \in Env \quad \text{sse}$$

$$\vdash d \rightarrow^* \rho$$

$$Elab(d_1; d) = \text{ambiente generato dall'applicazione delle regole della semantica dinamica}$$

Equivalenza : $\equiv \subseteq \text{Dec} \times \text{Dec}$

$$d_1 \equiv d_2 \quad \text{sse} \quad \text{Elab}(d_1) = \text{Elab}(d_2)$$

Esempio

$$\frac{\text{const } x: \text{int} = 2 \text{ in} \quad \left(\text{const } y: \text{int} = x+1; \text{const } z: \text{int} = y+x \right)}{d = d_2; d_3}$$

$\overbrace{\text{const } x: \text{int} = 2}^{d_1}$ $\overbrace{\text{const } z: \text{int} = y+x}^{d_3}$
 $\underbrace{\text{const } y: \text{int} = x+1}_{d_2}$

$$\boxed{d_1 \text{ in } d}$$

Primo passo:

$$\vdash d_1 \rightarrow^* ? \text{ ①}$$

$$\vdash d_1 \text{ in } d \rightarrow ? \text{ in } d$$

① $\vdash \text{const } x: \text{int} = 2 \rightarrow [x \mapsto 2]$ completa il passo 1

$$\vdash d_1 \rightarrow [x \mapsto 2]$$

$$\vdash d_1 \text{ in } d \xrightarrow{\text{①}} [x \mapsto 2] \text{ in } d$$

Primo passo di elaborazione

Secondo passo *

$$[x \mapsto 2] \vdash d \rightarrow^* ?$$

$$\vdash \underline{[x \mapsto 2] \text{ in } d} \rightarrow [x \mapsto 2] \text{ in } ?$$

Elaboriamo d : primo passo

$$d_2; d_3 = d$$

$$\frac{[x \mapsto 2] \vdash d_2 \xrightarrow{\text{②}}}{[x \mapsto 2] \vdash d_2; d_3 \rightarrow}$$

$$[x \mapsto 2] \vdash d_2; d_3 \rightarrow$$

Globalization d2

$$\frac{[x \mapsto 2] \vdash x+1 \rightarrow^* \textcircled{20}}{[x \mapsto 2] \text{ const } y: \text{int} = x+1 \rightarrow}$$

② Valutazione di $x+1$

$$\cdot \frac{\overbrace{[x \mapsto 2] \vdash x \rightarrow 2}^p}{[x \mapsto 2] \vdash x+1 \rightarrow 2+1} \quad \text{perché } p(x)=2$$

$$\cdot [x \mapsto 2] \vdash 2+1 \rightarrow 3 \quad \text{perché } 2+1=3$$

$$\Rightarrow [x \mapsto 2] \vdash x+1 \rightarrow^* 3 \quad \text{complete la valutazione } \textcircled{20}$$

$$\frac{[x \mapsto 2] \vdash x+1 \rightarrow^* 3}{[x \mapsto 2] \vdash \text{const } y: \text{int} = x+1 \rightarrow [y \mapsto 3]} \quad \text{complete } \textcircled{2}$$

$$\frac{[x \mapsto 2] \vdash d_2 \rightarrow [y \mapsto 3]}{[x \mapsto 2] \vdash d_2; d_3 \rightarrow [y \mapsto 3]; d_3}$$

Secondo passo di elaborazione di $d = d_2; d_3$

$$\textcircled{*} \frac{[x \mapsto 2, y \mapsto 3] \quad [x \mapsto 2][y \mapsto 3] \vdash d_3 \rightarrow \textcircled{3}}{[x \mapsto 2] \mapsto [y \mapsto 3]; d_3 \rightarrow}$$

③ Evoluzione di d_3

$$\textcircled{*} \frac{[x \mapsto 2, y \mapsto 3] \vdash y + x \rightarrow \textcircled{30}}{[x \mapsto 2, y \mapsto 3] \vdash \text{const } \underline{2} : \text{int} = y + x \rightarrow}$$

③a Valutazione di $y + x$

$$\left\{ \begin{array}{l} \frac{\overbrace{[x \mapsto 2, y \mapsto 3] \mapsto y}^p \rightarrow 3}{[x \mapsto 2, y \mapsto 3] \vdash y + x \rightarrow 3 + x} \quad \text{perché } p(y) = 3 \\ \frac{\overbrace{[x \mapsto 2, y \mapsto 3] \vdash x}^p \rightarrow 2}{[x \mapsto 2, y \mapsto 3] \vdash 3 + x \rightarrow 3 + 2} \quad \text{perché } p(x) = 2 \\ [x \mapsto 2, y \mapsto 3] \vdash 3 + 2 \rightarrow 5 \quad \text{perché } 3 + 2 = 5 \\ \rightarrow [x \mapsto 2, y \mapsto 3] \vdash y + x \rightarrow^* 5 \quad \text{completato il passo } \textcircled{30} \end{array} \right.$$

$$\textcircled{*} \frac{[x \mapsto 2, y \mapsto 3] \mapsto y + x \rightarrow^* 5}{[x \mapsto 2, y \mapsto 3] \vdash d_3 \rightarrow [x \mapsto 3]}$$

completato il passo ③

$$\textcircled{*} \quad \frac{[x \mapsto 2, y \mapsto 3] \vdash d_3 \rightarrow [z \mapsto 5]}{[x \mapsto 2] \vdash [y \mapsto 3]; d_3 \rightarrow [y \mapsto 3]; [z \mapsto 5]}$$

passo successivo:

$$[x \mapsto 2] \vdash [y \mapsto 3]; [z \mapsto 5] \rightarrow [y \mapsto 3, z \mapsto 5]$$

|| questo completa l'elaborazione di $d_2; d_3$

$$[x \mapsto 2] \vdash d \rightarrow^* [y \mapsto 3, z \mapsto 5] \quad \text{complete} \textcircled{a}$$

$$\frac{[x \mapsto 2] \vdash d \rightarrow^* [y \mapsto 3, z \mapsto 5]}{\vdash [x \mapsto 2] \text{ in } d \rightarrow [x \mapsto 2] \text{ in } [y \mapsto 3, z \mapsto 5]}$$

ultimo passo

$$\vdash [x \mapsto 2] \text{ in } [y \mapsto 3, z \mapsto 5] \rightarrow \underline{\underline{[y \mapsto 3, z \mapsto 5]}}$$

$$\Rightarrow \vdash d_1; d \rightarrow^* [y \mapsto 3, z \mapsto 5]$$

$$\equiv \text{Elab}(d_1; d) = [y \mapsto 3, z \mapsto 5]$$